

Citation-Level Unfaithfulness of Small Language Models on Vietnamese Banking Regulation: A Provenance-Guarded RAG Benchmark

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Abstract—Retrieval-augmented generation (RAG) is the standard remedy for hallucination in small language models, but its benefit is usually reported as an aggregate accuracy delta that hides where an answer came from. We present RegRAG-VN, a benchmark and evaluation framework for Vietnamese banking and payment regulation whose retrieval corpus is aligned with the questions by an exact-match coverage invariant. The benchmark holds 64 answerable questions drawn from State Bank of Vietnam circulars, decrees, and laws, alongside 24 unanswerable probes paired with an explicit abstention sentinel. We evaluate models of at most 7B parameters under closed-book, BM25, and dense retrieval, scoring article-level (*Điều*) and clause-level (*Khoản*) citation accuracy alongside abstention. Across a $3 \times 3 \times 88$ generation campaign on a single GPU, retrieval augmentation on a fixture whose passage pool contains every gold passage raises article-level citation F1 from 0.005 (closed-book) to 0.32–0.43. However, 43–49% of RAG answers on answerable questions still fail citation-level checking—about 34% even when the governing passage reaches the prompt intact—and models frequently emit refusal language before answering anyway; a provenance discipline tags corpus tiers and retriever backends throughout the pipeline so that degraded or unverified data cannot silently enter the results.

Index Terms—retrieval-augmented generation, hallucination, citation faithfulness, abstention, Vietnamese legal NLP, benchmark

I. INTRODUCTION

Regulatory compliance in Vietnam is a language problem with legal consequences. A bank officer who answers a customer’s question about payment-card limits, or a compliance analyst who checks whether a new product violates a circular of the State Bank of Vietnam (SBV), must locate the governing provision among hundreds of circulars (*Thông tư*), decrees (*Nghị định*) and laws (*Lệnh*), each organised into numbered articles (*Điều*) and clauses (*Khoản*). Language models are an obvious aid, but the models that regional banks and fintechs can actually deploy are small: a 3–7B parameter model served on a single commodity GPU, not a frontier API whose data leaves the country. Whether such models answer Vietnamese regulatory questions faithfully and cite the governing provisions accurately is therefore a practical deployment question as well as a research one.

Retrieval-augmented generation is well established as a remedy for hallucination in English knowledge-intensive tasks [1], [2], and simply replicating that aggregate gain in Vietnamese provides limited practical insight. The failure mode that aggregate accuracy conceals is *citation-level unfaithfulness*: a model produces a fluent, plausible answer that carries the wrong *Điều* or *Khoản*, or cites an instrument that does not govern the question at all, instead of abstaining. For a compliance tool, a wrong answer with a confident, wrong citation is more dangerous than an obviously wrong answer, because the citation is what a human reviewer checks first. Measuring that behaviour requires a benchmark in which the correct citation is known per question and in which the corpus provably contains the evidence.

RegRAG-VN is designed around this requirement. Its corpus is *QA-driven*: the legal instruments ingested are exactly those cited by the benchmark questions, and a coverage invariant requires every retained question’s verbatim gold passage to match the ingested text exactly at chunk level. Ground truth is therefore verifiably present in the retrieval corpus rather than assumed. On top of this corpus we measure three research questions:

- **RQ1 (hallucination reduction)**. How much does RAG reduce hallucination versus closed-book answering for small models ($\leq 7B$) on Vietnamese regulatory questions?
- **RQ2 (citation and abstention)**. How accurately do models cite the supporting article (*Điều*) and clause (*Khoản*), and do they abstain when the corpus does not contain the answer?
- **RQ3 (sparse versus dense)**. How does BM25 with Vietnamese compound-word segmentation compare with multilingual dense embeddings (BGE-M3) on legal retrieval?

A benchmark of this nature is only as reliable as the pipeline that produces its empirical findings. In multi-component RAG evaluation, silent software fallbacks—such as substituting mock similarity rankings when an optional neural library is missing, or reverting to unsegmented whitespace tokenization—can fabricate plausible numerical results with-

out warning. To ensure strict scientific reproducibility, we treat provenance tracking as a fundamental methodological requirement: every chunk, retrieved candidate, generation output, and evaluation record carries immutable metadata specifying its corpus source and active backend. Our evaluation harness explicitly rejects any row carrying degraded or unverified execution tags; Section IV details this discipline.

The contributions of this paper are: (i) RegRAG-VN, a QA-driven Vietnamese banking regulation benchmark featuring 64 answerable statutory questions and 24 unanswerable operational probes, governed by a coverage invariant ensuring supporting evidence is verifiably present; (ii) an evaluation methodology that scores legal attribution at article (*Điều*) and clause (*Khoản*) granularity while assessing model abstention under coverage gaps; and (iii) a provenance architecture for RAG benchmarks that tags and gates execution integrity across complex multi-stage pipelines.

II. RELATED WORK

RAG and its evaluation. Retrieval-augmented generation conditions a generator on passages retrieved from a corpus [1], with dense passage retrieval [3] and sparse BM25 scoring [4] as the two dominant retriever families [2]. General RAG suites evaluate component capabilities such as noise robustness and negative rejection [5], and reference-free frameworks estimate faithfulness and answer relevance [6], but these suites are English-centric and none scores the correctness of a statutory citation, the unit a compliance reviewer trusts. Two legal-domain benchmarks move toward that requirement. LegalBench-RAG scores retrieval against gold supporting spans so that generation quality cannot mask evidence-selection errors [7], and Legal RAG Bench evaluates end-to-end legal RAG with a hierarchical error decomposition [8]. We report three failure classes accordingly: retrieval failure, where no gold provision reaches the prompt; generation failure, where the provision is present but the answer asserts unsupported claims; and citation mismatch, where the answer names a wrong or non-governing provision.

Hallucination and faithfulness measurement. Hallucination in natural language generation is surveyed along the intrinsic/extrinsic axis [9]. In summarisation, faithfulness was operationalised by evaluating whether source text entails summary facts [10], while atomic-fact decomposition enabled precision scoring per claim [11]. Attribution research further formalises the requirement that generated statements be supported by cited sources, scoring citation quality separately from answer correctness [12]. These risks are documented in deployment: an audit of leading legal research platforms found answer, source, citation, and completeness failures on every tool tested [13]. Our citation metric follows this attribution line but is stricter in one respect: the cited unit is a statutory article or clause, so a citation to the right instrument but the wrong article scores zero, matching how a legal reviewer reads a reference.

Legal-domain language models. Legal NLP benchmarks such as LexGLUE establish that legal text behaves differently

from general text [14]. Vietnamese legal text adds a structural property that most legal-RAG work ignores: provisions are numbered hierarchically (*Điều* within *Chương*, *Khoản* within *Điều*), so the chunking granularity is itself an experimental variable. Prior work demonstrates that mixing chunk granularities outperforms any single fixed granularity for RAG [15]; our *Điều/Khoản* segmentation adapts this principle directly to the statutory hierarchy. Long-context dynamics matter as well, since retrieved passages compete for attention within the prompt [16].

Vietnamese legal NLP. VLQA links legal questions to answers and evidence documents [17], VLegal-Bench evaluates multi-task legal reasoning [18], and ViLegalLM adapts language models to Vietnamese legal text [19]. Nguyễn Bá et al. compare lexical, dense, and hybrid retrieval for Vietnamese legal QA [20]. Within these resources we did not find one that couples article-level citation scoring and unanswerable probes with a corpus whose coverage of the governing provisions is verified; that conjunction is the one RegRAG-VN occupies. The sparse arm depends on Vietnamese word segmentation [21]: because the orthography places spaces between syllables rather than words, a whitespace tokenizer splits compound legal terms such as *ngân hàng* or *thẻ tín dụng* into fragmented units, so we segment queries and chunks before BM25 scoring (RQ3).

III. THE BENCHMARK

A. Domain and QA-driven corpus construction

The domain is Vietnamese banking and payment regulation: SBV circulars on payment services, cards, non-cash payment and credit-institution organisation, together with the governing decrees and laws. Rather than querying a broad collection of legal instruments, the retrieval corpus is derived directly from the benchmark questions. Each gold question was authored together with a verbatim passage from the instrument that answers it and the source URL of that instrument. Instrument identifiers are resolved from the URL through a human-verified manifest; where a URL is opaque, the row is kept with an explicit UNRESOLVED: identifier and reported, never dropped, so the benchmark cannot silently shrink. When a passage cites multiple candidate instruments, the named instrument is primary; spreadsheet URL ordering provides no priority.

The current benchmark holds 88 rows: 64 answerable questions, each with a gold passage recording the supporting provision and at least one source URL, and 24 unanswerable probes. Because gold passages reflect annotator records and may not represent verbatim extracts, the pipeline verifies each passage against the corpus at ingestion. When a recorded passage is a summary rather than an exact extract, the verbatim article text is substituted from the source instrument and logged. In the delivered revision, 63 of the 64 passages consist of verbatim article text substituted under a guarded repair procedure, which rewrites a row only if the cited instrument is present in the corpus, the cited article exists, and the annotator summary clears a lexical overlap threshold against that article;

rows whose cited instrument is absent from the corpus are excluded from the answerable set. Of the remaining rows, one cites a *Phụ lục* annex rather than a numbered article and is retained as an annotator summary; a further row, whose governing instrument is absent, was moved to the probe arm and lies outside the 64. Since substituted passages originate directly from the ingested corpus, verbatim string coverage on repaired rows is partly self-satisfying; we therefore report it as an internal data-quality metric and anchor formal evaluation on article-level citation accuracy. Article-level gold citations (*Điều*) are derivable for 63 of the 64 answerable rows; clause-level gold (*Khoản*) is present for 40. We therefore score citation accuracy at article level and treat clause level as optional: a clause reference is credited when present and correct but is never required for a match, because requiring it would penalise correct answers for questions whose governing provision has no clause structure.

B. Two-tier corpus and the coverage invariant

The corpus has two tiers with identical interfaces. Tier 1 is a development fixture of the verbatim gold passages, one chunk per passage, built so that indexing, retrieval, prompting, generation and scoring could be integrated before the real instruments arrived. Although each question’s gold passage is in Tier 1 by construction, top-3 retrieval over the 64-chunk fixture is not guaranteed to surface it—the gold chunk reaches the top 3 for only 33 of 64 answerable questions under BM25 and 47 of 64 under dense retrieval—so Tier 1 measures generation under a mixture of evidence conditions. Tier 1 is therefore never treated as corpus evidence, and the generation campaign that used it runs under the disclosure of §V-A. Tier 2 is the segmented full text of the ingested instruments and is the corpus on which retrieval quality is measured (Table III). Segmentation follows the statutory hierarchy: articles (*Điều*) are chunked at clause (*Khoản*) boundaries—one chunk per clause, short articles single chunks—and preambles are retained rather than discarded.

The coverage invariant enforces alignment between the tiers: every retained answerable question’s verbatim gold passage must match a Tier 2 chunk exactly under strict normalisation, in either direction over at least 30 characters—for clause-chunked articles the match is the governing clause inside the passage—with the single annex-citing row of §III-A excepted (Unicode NFC plus whitespace collapse, diacritics preserved). A passage matching only after diacritic folding is classified as extraction damage, since lost diacritics would corrupt retrieval and citation scoring. A passage matching only after whitespace removal is logged as a second diagnostic category, an artifact of gazette PDFs that insert spurious intra-word spaces. Neither type of degradation is accepted as an exact match. Any question whose governing instrument could not be obtained is excluded from the answerable set; retaining it would falsely penalise models for hallucinating evidence that was never retrievable.

C. Unanswerable probes

The 24 probes evaluate the abstention component of RQ2 using questions deliberately unanswerable from the corpus; their gold target is the sentinel string *Không có trong kho văn bản* (“not in the document corpus”). These scenarios represent plausible, in-domain banking situations whose governing provisions lie outside the repository. Examples include opening payment accounts for condominium management boards, kindergartens, and foreign trade-representative offices; collateral held by a spouse or an owner under 18; and the reporting frequency of the capital-adequacy ratio. The capital-adequacy question was originally annotated as answerable, but source verification revealed it is governed by an uningested statistical-reporting circular, prompting its reassignment to the probe arm.

By testing realistic questions governed by missing instruments, the probes evaluate abstention under a corpus-coverage gap rather than simple rejection of out-of-domain topics. This setting is intentionally demanding: abstention cannot be triggered by topical distance but requires recognising that the retrieved evidence does not support an answer. We state as a limitation that the probe set contains no precision-trap and no cross-jurisdictional arm, so abstention on numerically near-miss questions and on foreign rules is not measured. A model that answers a probe fluently has failed abstention even if the answer is globally true, because the benchmark strictly evaluates what the corpus supports.

D. Annotation and inter-annotator reliability

Questions were authored by two domain annotators and verified against their source instruments at ingestion (§III-A); wrong labels were corrected from documented findings rather than scored as-is. Inter-annotator agreement on the gold labels was not measured, and we state the resulting author-bias risk as a limitation.

IV. METHOD

A. Retrieval

We compare three retrieval modes at top- $k = 3$. Under *closed-book*, the prompt contains only the question without context. For *RAG-BM25*, queries and corpus chunks are segmented with PyVi [22] to preserve compound legal terms, followed by BM25 scoring [4], [23]. For *RAG-Dense*, queries and chunks are embedded using multilingual BGE-M3 [24] and ranked by cosine similarity. In both RAG configurations, the top-3 retrieved passages are concatenated into the prompt with identical formatting under a shared 7,000-character context budget (447 of 528 RAG prompts are clipped and 240 ranked passages dropped), so retrieval representation is the only factor varied by design, though the budget’s clipping depends on each retriever’s ranking.

B. Generation

We evaluate three instruction-tuned models of at most 7B parameters: Qwen2.5-7B-Instruct and Qwen2.5-3B-Instruct [25], alongside the Vietnamese-focused Vistral-7B-Chat [26]

as an architecturally distinct third model. All models are served in 4-bit quantisation (NF4 weights, float16 compute) on a single 24 GB GPU. Each model passes an automated memory preflight, and chat-template application is verified on sample outputs. Every prompt includes explicit instructions specifying a refusal keyphrase, ensuring abstention is measured as a deliberate behavior rather than an omission.

C. Evaluation metrics

For each evaluation instance, we compute four core metrics. Groundedness assesses whether atomic assertions in the generated answer are entailed by the retrieved context, following claim-level verification methodologies [11], [12], and the hallucination rate identifies statements that contradict statutory text or introduce unsupported assertions. Citation precision, recall, and F1 are scored at the article (*Điều*) level, with optional credit for clause (*Khoản*) accuracy (citations to the correct article number under an incorrect legal instrument score zero). Abstention accuracy on unanswerable probes measures whether the model emits the instructed refusal keyphrase without generating substantive claims.

Our automated scoring pipeline is deterministic, lexical, and released in full without manual adjustment. We validated it against 50 expert-annotated responses stratified across models and modes. Every correctness and abstention call the scorer makes is confirmed by the expert (3/3 and 10/10), but it misses 15 expert-correct answers and 8 expert-judged abstentions, so those positive calls are exact and those rates conservative. Scorer-expert agreement is $\kappa = 0.20$ for correctness, 0.39 for hallucination, and 0.62 for abstention [27]; on hallucination the scorer misses ten and over-flags five, netting the undercount of five that makes the rates in Table I mild underestimates. The validation post-dates the campaign: Tables I and II were aggregated with the scorer still unvalidated (`strict=False` in the log), and the κ values above are the completed check.

D. Provenance discipline

Every chunk, retrieved result, generation, and evaluation record carries two provenance tags: `corpus_source` (`tier1_passages` or `tier2_full`) and `retriever_backend` (e.g., `bm25-rank_bm25+pyvi` or `BAAI/bge-m3`, or `DEGRADED:<reason>`). Aggregation functions are built to refuse rows whose backend is unverified, degraded, or mismatched; no row carried a degraded tag in the campaign, but the tables of §V were aggregated with this gate relaxed pending the scorer validation (§IV-C).

This architecture was developed in response to concrete vulnerabilities in modular RAG pipelines. First, unhandled import exceptions can let mock dense-retrieval classes return corpus-ordered chunks with synthetic scores, fabricating a whole experimental condition without failure. Second, sparse retrievers can silently fall back from compound-word segmentation to naive whitespace splitting, undermining lexical evaluation while still self-reporting as BM25. Finally, corpus validation quarantined an early synthetic seed file with non-contiguous articles and missing administrative headers, as

fabricated provisions act as corrupt distractors. The rejection path did fire at ingest level: a Tier-2 re-ingest was rejected on duplicate chunk identifiers, which is why generation ran on the Tier-1 fixture (§V-A) while Table III used the validated Tier-2 build. By enforcing immutable provenance metadata across every stage, RegRAG-VN keeps experimental conclusions anchored to authentic documents and disclosed execution paths.

V. EXPERIMENTAL DESIGN AND RESULTS

We evaluate three models across three retrieval modes (closed-book, RAG-BM25, and RAG-Dense) at $\text{top-}k = 3$ over all 88 benchmark rows, producing 792 generated answers through the pipeline of Fig. 1. The campaign ran in one ten-hour window on a single RTX 3090 (24 GB), serving each model in 4-bit NF4 quantisation with greedy decoding (temperature 0, seed 1234, at most 512 new tokens) and one model resident at a time. All 792 prompts round-trip byte-identical through the harness; 709 answers finish naturally and 83 hit the token cap and are analysed as generated. With 64 answerable rows and 24 probes per cell, binomial noise is on the order of six and ten percentage points respectively, so we report point estimates without significance tests and read small differences between cells cautiously. Retrieval quality is reported separately (Table III), because a RAG arm whose retriever never finds the gold passage measures prompting rather than retrieval.

A. Evidence Conditioning and Scope

The generation experiments run on the curated Tier-1 fixture, in which every answerable question’s supporting statutory passage is present by design. The guarantee holds at the pool level, not the prompt level (§III-B): the results below mix generation-side citation errors with fixture-retrieval misses, and RQ2 splits the unfaithfulness rates by evidence condition. Three implications follow:

- **Upper-Bound Conditioning:** Tables I and II measure generation on a fixture whose pool contains every gold passage; the gains over closed-book baselines are therefore an empirical upper bound on retrieval-augmented performance.
- **Arm Differences:** differences between RAG-BM25 and RAG-Dense reflect both retrieval behaviour on the fixture and prompt composition; the two configurations yield identical answers in only 45 of 264 shared instances.
- **Independent Retrieval Evaluation:** Retrieval performance is evaluated independently on the 3,703-chunk Tier-2 corpus (Table III) across 22 instruments.

B. RQ1: hallucination reductions partly reflect a shift to refusal

Closed-book hallucination rates span 0.44–0.75 (Table I). On all answerable rows, conditioning on retrieved evidence lowers the rate for both Qwen models (Qwen2.5-7B 0.438 $\rightarrow$ 0.344/0.313, Qwen2.5-3B 0.750 $\rightarrow$ 0.469/0.422 under BM25/dense). Restricting the comparison to rows each model

TABLE I

HALLUCINATION, GROUNDEDNESS AND ABSTENTION BY MODEL AND RETRIEVAL MODE (RQ1, RQ2-ABSTENTION). HALLUC. = HALLUCINATION RATE OVER THE 64 ANSWERABLE QUESTIONS; GROUND. = MEAN SENTENCE-LEVEL GROUNDEDNESS OVER ROWS WITH ASSESSABLE RETRIEVED CONTEXT (CLOSED-BOOK HAS NONE, HENCE –); ABST. = SHARE OF THE 24 PROBES CORRECTLY REFUSED.

Model	Mode	Halluc. ↓	Ground. ↑	Abst. ↑
Qwen2.5-7B	closed-book	0.438	–	0.042
Qwen2.5-7B	RAG-BM25	0.344	0.985	0.500
Qwen2.5-7B	RAG-Dense	0.313	0.951	0.375
Qwen2.5-3B	closed-book	0.750	–	0.000
Qwen2.5-3B	RAG-BM25	0.469	0.950	0.000
Qwen2.5-3B	RAG-Dense	0.422	0.916	0.125
Vistral-7B	closed-book	0.625	–	0.000
Vistral-7B	RAG-BM25	0.672	0.912	0.000
Vistral-7B	RAG-Dense	0.594	0.939	0.000

TABLE II

CITATION ACCURACY AT ARTICLE (*Điều*) LEVEL AND OPTIONAL CLAUSE (*Khoản*) LEVEL, POOLED OVER THE THREE MODELS (RQ2). P AND R ARE AVERAGED PER ROW OVER THE 189 SCORABLE ANSWERABLE ROWS (63 QUESTIONS $\times$ 3 MODELS; ONE ANNEX-CITING QUESTION HAS NO ARTICLE GOLD); A CORRECT ARTICLE IN THE WRONG INSTRUMENT SCORES ZERO. *Khoản* ACCURACY IS OVER THE 120 ROWS WHOSE GOLD CITATION NAMES A CLAUSE.

Mode	<i>Điều</i> P / R / F1			<i>Khoản</i> acc.
closed-book	0.005	0.005	0.005	0.008
RAG-BM25	0.293	0.354	0.321	0.300
RAG-Dense	0.399	0.476	0.434	0.317

actually answered removes most of the 7B effect: answered-row rates are $0.438 \rightarrow 0.458/0.408$ for Qwen2.5-7B, indistinguishable from closed-book at this sample size, whereas Qwen2.5-3B keeps a reduction ($0.762 \rightarrow 0.556/0.450$), unambiguous only in the dense arm. In contrast, Vistral-7B exhibits no reduction; its BM25 rate is marginally *higher* than its closed-book baseline (0.672 vs. 0.625) because it paraphrases retrieved text into unsupported assertions rather than extracting it. Groundedness is high whenever context is present (0.91–0.98). The aggregate delta therefore entangles with refusal (a refused row cannot be scored hallucinated) and masks an opposing calibration shift: Qwen2.5-7B falsely refuses 25.0% and 23.4% of answerable questions under RAG against 0% closed-book, and Qwen2.5-3B falsely refuses 15.6% and 6.3%; 21 of the BM25 arm’s 28 answerable refusals occur where the governing text did not reach the prompt.

C. RQ2: citation-level unfaithfulness is the dominant failure

Closed-book citation performance is near zero: article-level F1 = 0.005, with 102 of 192 closed-book answerable rows (64×3 models) asserting a provision that does not govern the question and 76 citing an instrument other than the governing one (categories overlap). RAG lifts citation F1 to 0.321 (BM25) and 0.434 (dense), but citation-level unfaithfulness remains the modal failure: 49% (BM25) and 43% (dense) of answerable rows are citation-unfaithful—wrong instrument, non-covering provision, numeric-threshold mismatch, or an

TABLE III

RETRIEVAL QUALITY ON THE TIER 2 CORPUS (RQ3). RECALL AT INSTRUMENT (DOC) AND ARTICLE (*Điều*) LEVEL, $k=1/3$, OVER THE ANSWERABLE SET (DOC $n=64$; ARTICLE $n=63$, ONE ROW CITES AN ANNEX).

Retriever	Doc@1	Doc@3	<i>Điều</i> @1	<i>Điều</i> @3
BM25 (pyvi)	0.734	0.906	0.381	0.492
BGE-M3 dense	0.719	0.891	0.365	0.524

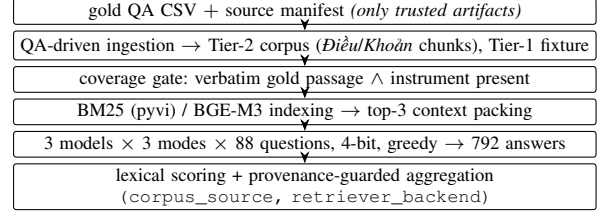


Fig. 1. The RegRAG-VN pipeline. Every record crossing a component boundary carries the `corpus_source` and `retriever_backend` tags; the coverage gate excludes a question rather than mislabel it as hallucinated.

uncited assertion; wrong or non-covering citations alone account for 41% and 34%. Splitting on whether the governing text actually reached the prompt, the unfaithfulness rate falls to about one-third in both arms when it arrives intact (19/57 BM25, 29/84 dense), moves to 33–44% when the context budget clips it, and rises to 60–67% when it is absent: unfaithfulness persists under full evidence conditioning and roughly doubles without it ($\chi^2=10.5/13.6$, intact vs. absent; the split is selected by the retriever’s own ranking). Pooled clause-level (*Khoản*) accuracy never exceeds 0.317 (Table II). Under closed-book evaluation, 71 of 72 probe rows receive substantive answers despite the keyphrase instruction. Another 80 of the 192 closed-book answerable rows open with refusal language yet answer anyway (65 reproduce the keyphrase verbatim, 15 only paraphrase it); a keyphrase scorer would credit the verbatim kind as correct refusals, motivating our scorer’s assertion requirement (§IV-C); across all configurations, Vistral-7B refuses no probe in any mode.

Qualitative inspection of model answers reveals the underlying failure patterns. In question Q006 (closed-book), where the governing provision is Article 52(4) of the 2024 Notarization Law, Qwen2.5-7B instead cites “*Điều 405 Bộ luật Dân sự 2015*” and appends a statutory quotation that appears nowhere in the ingested corpus. In Q008 (closed-book), the model emits the instructed keyphrase but immediately follows with a substantive explanation citing “*Nghị định 02/2021/NĐ-CP... Điều 14, Điều 15*”, demonstrating the hedge-then-answer pattern on an irrelevant decree. Returning to Q006 under RAG-BM25 (Vistral-7B), with that gold text intact in the prompt, the answer is substantively correct yet opens by citing “*Điều 52 của Nghị định 23/2015/NĐ-CP, sửa đổi bởi Nghị định 111/2020/NĐ-CP...*”—the right article number under the wrong instrument, and later cites the governing instrument under a wrong article (*Điều 32*), with the governing evidence verbatim in the prompt. In regulatory compliance, such citation

errors are particularly hazardous because human auditors use citations as their primary index for verification.

D. RQ3: document-level retrieval is strong, article-level is not

On the Tier-2 corpus (Table III), BM25 with Vietnamese segmentation retrieves a chunk of the cited instrument for 90.6% of answerable questions at $k=3$ (73.4% at $k=1$) but places the cited *Điều* itself in the top 3 for only 49.2% (38.1% at $k=1$). The dense arm is close behind at document level (89.1% at $k=3$) and edges BM25 at article level (52.4% against 49.2%)—a two-question difference, 33 versus 31 of 63, that this sample size cannot resolve. Document-level retrieval is strong but incomplete, with six (BM25) and seven (dense) of 64 questions retrieving no chunk of the governing instrument at $k=3$, and placing the specific governing article in context remains difficult: unless a model cites correctly from parametric memory, only about half of questions can carry the governing article into the prompt, capping end-to-end citation well below the conditioned rates of RQ2.

VI. CONCLUSION

RegRAG-VN establishes an empirical evaluation framework for regulatory RAG by coupling benchmark queries directly to statutory provisions, scoring citations at article and clause resolution, and assessing abstention under corpus coverage gaps. Retrieval augmentation shifts small models from answering toward refusal, and the raw hallucination reduction survives that correction only for the smallest model; citation fidelity remains the critical vulnerability, with 43–49% of RAG answers on answerable questions failing citation-level checking, a rate near 34% even when the source provision reaches the prompt intact, pooled clause accuracy at or below 0.32, and brittle refusal calibration throughout. Retrieval experiments on the full 3,703-chunk statutory corpus show that document-level recall at $k=3$ reaches 90% while article-level recall drops below 53%. By formalizing a provenance-guarded evaluation architecture, we provide a verifiable foundation for future research on faithful legal language models.

DATA AND CODE AVAILABILITY

The RegRAG-VN benchmark, pipelines, evaluation suites, and full experimental logs are available in our public repository, with a permanent Zenodo archive and DOI established at submission. Curated gold questions, document manifests, and coverage reports are the authoritative artifacts from which all derived assets can be reproduced; ingested legal texts are redistributed unmodified from official Vietnamese government publications.

REFERENCES

- [1] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Adv. Neural Inf. Process. Syst.*, 2020.
- [2] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, and H. Wang, “Retrieval-augmented generation for large language models: A survey,” *arXiv preprint arXiv:2312.10997*, 2023.
- [3] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. Yih, “Dense passage retrieval for open-domain question answering,” in *Proc. EMNLP*, 2020.
- [4] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Found. Trends Inf. Retr.*, vol. 3, no. 4, pp. 333–389, 2009.
- [5] J. Liu, Q. Wang, Y. Wang, Y. Teng, X. Qiu, J. Song, M. Zhang, and Y. Xiao, “Benchmarking large language models in retrieval-augmented generation,” in *Proc. AAAI*, 2024.
- [6] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAS: Automated evaluation of retrieval augmented generation,” in *Proc. EACL*, 2024.
- [7] N. Pipitone and G. Houir Alami, “LegalBench-RAG: A benchmark for retrieval-augmented generation in the legal domain,” *arXiv preprint arXiv:2408.10343*, 2024.
- [8] A.-R. Butler and U. Butler, “Legal RAG bench: An end-to-end benchmark for legal RAG,” *arXiv preprint arXiv:2603.01710*, 2026.
- [9] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, “Survey of hallucination in natural language generation,” *ACM Comput. Surv.*, vol. 55, no. 12, 2023.
- [10] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in *Proc. ACL*, 2020.
- [11] S. Min, K. Krishna, X. Lyu, M. Lewis, W. Yih, P. W. Koh, M. Iyyer, L. Zettlemoyer, and H. Hajishirzi, “FactScore: Fine-grained atomic evaluation of factual precision in long form text generation,” in *Proc. EMNLP*, 2023.
- [12] H. Rashkin, V. Nikolaev, M. Lamm, L. Aroyo, M. Collins, D. Das, S. Petrov, G. S. Tomar, I. Turc, and D. Reitter, “Measuring attribution in natural language generation models,” *Computational Linguistics*, vol. 49, no. 4, 2023.
- [13] V. Magesh, F. Surani, M. Dahl, M. Suzgun, C. D. Manning, and D. E. Ho, “Hallucination-free? Assessing the reliability of leading AI legal research tools,” *Journal of Empirical Legal Studies*, vol. 22, no. 2, pp. 216–242, 2025.
- [14] I. Chalkidis, M. Fergadiotis, P. Malakasiotis, N. Aletras, and I. Androutsopoulos, “LexGLUE: A benchmark dataset for legal language understanding in English,” in *Proc. ACL*, 2022.
- [15] Z. Zhong, H. Liu, X. Cui, X. Zhang, and Z. Qin, “Mix-of-granularity: Optimize the chunking granularity for retrieval-augmented generation,” in *Proc. COLING*, 2025.
- [16] N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang, “Lost in the middle: How language models use long contexts,” *Trans. Assoc. Comput. Linguist.*, vol. 12, 2024.
- [17] T.-M. Nguyen, H.-T. Nguyen, T.-K. Dao, X.-H. Phan, H.-T. Nguyen, and T.-H.-Y. Vuong, “VLQA: The first comprehensive, large, and high-quality Vietnamese dataset for legal question answering,” *arXiv preprint arXiv:2507.19995*, 2025.
- [18] T.-D. Nguyen, M.-A. Nguyen, T.-D. Hoang *et al.*, “VLegal-Bench: A cognitively grounded benchmark for Vietnamese legal reasoning of large language models,” *arXiv preprint arXiv:2512.14554*, 2025.
- [19] T.-P. Nguyen, Q.-N. Nguyen, and M.-T. Nguyen, “ViLegalLM: Language models for Vietnamese legal text,” in *Findings of ACL 2026*, 2026, pp. 36 136–36 150.
- [20] T. Nguyen Ba, V. Doan The, T. Pham Quang, and T. Tran Van, “Vietnamese legal information retrieval in question-answering system,” *arXiv preprint arXiv:2409.13699*, 2024.
- [21] T. Vu, D. Q. Nguyen, D. Q. Nguyen, M. Dras, and M. Johnson, “VnCoreNLP: A Vietnamese natural language processing toolkit,” in *Proc. NAACL-HLT Demonstrations*, 2018.
- [22] T. T. Nguyen, “PyVi: Vietnamese natural language processing toolkit,” <https://github.com/trungtin/pyvi>, accessed 2026-09-11.
- [23] D. Brown, “rank_bm25: BM25 algorithms in Python,” https://github.com/dorianbrown/rank_bm25, accessed 2026-09-11.
- [24] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” *arXiv preprint arXiv:2402.03216*, 2024.
- [25] A. Yang, B. Yang, B. Zhang *et al.*, “Qwen2.5 technical report,” *arXiv preprint arXiv:2412.15115*, 2024.
- [26] C. V. Nguyen, T. Nguyen, Q. Nguyen *et al.*, “Vistral-7B-Chat: Towards a state-of-the-art large language model for Vietnamese,” <https://huggingface.co/Viet-Mistral/Vistral-7B-Chat>, 2023.
- [27] J. Cohen, “A coefficient of agreement for nominal scales,” *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, 1960.